

Towards Human-Level Adaptation, Reasoning, and Efficient Spatial Intelligence

Shibo Zhao, Ph.D. Candidate in Computer Science, Robotics Institute, CMU

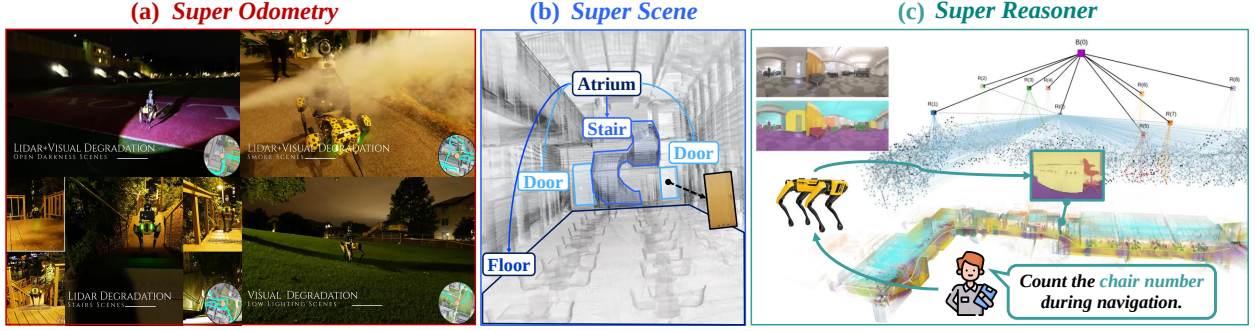


Figure 1: **Towards Real-time, Human-level Spatial Intelligence.** To enable scalable spatial intelligence, we propose: (a) **Super Odometry**: Real-time ego-motion estimation via adaptive fusion of visual, LiDAR, and inertial sensors; (b) **Super Scene**: Spatial-temporal 4D scene reconstruction for spatial reasoning; (c) **Super Reasoner**: Physical world understanding for efficient and intelligent autonomy.

Introduction

The development of autonomous systems capable of human-level reasoning, comprehension, and interaction within complex environments remains a significant challenge in robotics and artificial intelligence. While advancements in Large Language Models (LLMs), such as ChatGPT, and large-scale world modeling frameworks like Genie have demonstrated promising progress toward achieving Artificial General Intelligence (AGI), these approaches remain insufficient for robotics applications. Their limitations stem from an inability to address critical real-world tasks like reliable state estimation, rich environmental representation, and spatial reasoning, essential for robots to understand and interact effectively with the physical world.

To address these challenges, we believe robots need a comprehensive framework capable of reliably adapting to any environment, flexibly representing surroundings with varying levels of detail, and reasoning about physics efficiently. These capabilities are essential for achieving long-term autonomy and enabling robots to exhibit behavior approaching human-level performance. To this end, we introduce the concept of *Spatial Intelligence* (see Fig. 1), which incorporates a *human-level adaptive sensor fusion* framework for robust state estimation, a *4D scene representation* with flexible levels of detail, and *scene reasoning* to support autonomy.

Research Problems

- 1 Human-Level Sensing Adaptation:** The primary challenge of current robotic systems is their inability to adapt effectively to various environments, threatening the safety of robots. Our team addresses this limitation by introducing a hierarchical adaptation framework (**Super Odometry**) to seamlessly adjust different sensing modalities, which strikes a balance between redundancy and operational efficiency.
- 2 Spatio-Temporal 4D Reconstruction:** A major limitation of current scene reconstruction pipelines is that they generate a world model based on a single snapshot in time. However, robots need the ability to reason about the state of scenes at previous moments and predict how these scenes may evolve in the future. To address this challenge, our team proposes the development of spatial-temporal 4D reconstruction (**Super Scene**), which incrementally builds a world model over time. This approach not only enables detailed, photorealistic reconstructions but also facilitates abstract object-level mapping.
- 3 Human-Level Scene Reasoning:** The primary objective is to discover, learn, and transfer spatial common sense by analyzing the above 4D reconstruction with symbolic (physical, logical, and geometrical) reasoning abilities. Our approach (**Super Reasoner**) allows robots to have an integrated perception-plan-act pipeline by combining the powerful expressiveness of 4D model with explainable and rule-based reasoning. Such an integration will help robots comprehend physical relationships in their surroundings and make informed decisions based on these learned physical and logical rules.

Relation to Prior Work and State of the Art

Human-Level Sensing Adaptation: Real-time robotic autonomy relies on robust sensor fusion to navigate dynamic and degraded environments effectively. Towards stronger robustness, researchers adopted multi-modal sensor fusion, i.e., loosely-coupled [1] and tightly-coupled methods [2]. While these approaches have greatly advanced the field, none of the existing methods could overcome all degraded cases including visual degradation, geometrical degradation, and their mixed degradation [3]. Moreover, current methods struggle when multiple sensors simultaneously experience degradation or operate under off-nominal conditions for extended periods. Most approaches enhance robustness by fusing data from all available sensors, thereby ensuring redundancy and improving safety. However, this often leads to computationally intensive optimization, which can hinder real-time performance. To solve this problem, we propose *Super Odometry*, a novel framework with a hierarchical adaptation strategy. This approach dynamically adjusts to degradation across a spectrum, from mild to severe, and scale its responses to maintain efficiency and robustness.

Spatio-Temporal 4D Reconstruction: Compared to traditional 3D reconstruction methods, which focus on the static spatial structure of a scene, spatial-temporal (4D) reconstruction models dynamic, object-centric changes over time. This enables the capture of both evolving spatial configurations and temporal interactions within a scene [4] [5]. Building on this, we introduce the *Super Scene* representation, which aggregates spatial and temporal data at a higher level of abstraction. This approach is particularly valuable for Physical AI systems, where accurate digital counterparts allow for safe and proactive interaction with the real world. *Super Scene* representation enables robots to refine their behaviors through continuous learning in ever-changing environments by providing an adaptive, on-demand 4D reconstruction of the scene.

Human-Level Scene Reasoning: Achieving full spatial autonomy requires robots to reason over historical spatial-temporal observations and make informed decisions. Recent work has focused on understanding and reasoning about visual spatial relationships through implicit latent knowledge in large-scale visual-language models [6, 7]. In parallel, other approaches have explored explicit 3D scene representations, such as scene graphs, to facilitate spatial reasoning over pre-generated concepts [8] within an established framework. However, efficient understanding and reasoning about spatial relationships in dynamic environments remains under-explored, particularly in robotic applications, which require proactive and efficient decision-making. To address this gap, we introduce *Super Reasoner*, a human-level scene reasoning module that advances online spatial reasoning on the spatial-temporal 4D representation constructed by *Super Scene*, adaptively focusing on moving, manipulated, and emerging components to balance efficiency and accuracy.

Super Spatial Intelligence Framework

Fig. 2 illustrates the *Super Spatial Intelligence* framework, which comprises three key modules: *Super Odometry*, *Super Scene*, and *Super Reasoner*. The *Super Odometry* module ensures robust localization across diverse environments. The robot’s state is then transmitted to the *Super Scene* module, which reconstructs the surrounding environment in a temporal-spatial context. Meanwhile, the *Super Reasoner* module stores the robot’s state and the 4D scene representation to facilitate understanding. Together, these modules deliver expressiveness, efficiency, and reliable representations for the perception-planning-action pipeline, enabling a wide range of robotic tasks.

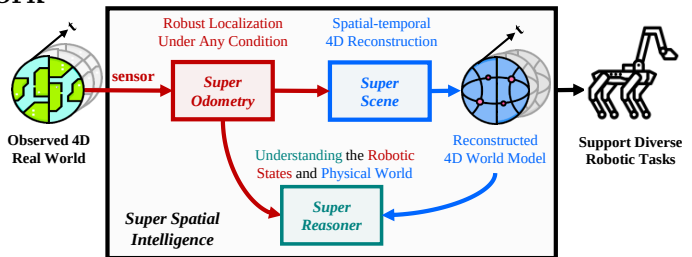


Figure 2: Pipeline overview of *Super Spatial Intelligence*.

Thrust #1: Embracing Human-level Sensing Adaptation Towards All-weather Environments.

The core challenge addressed by *Super Odometry* is maintaining high precision in ego-motion estimation under varying environmental conditions. We propose a hierarchical adaptation framework that dynamically reconfigures sensor fusion processes to respond to challenges such as occlusions, sensor degradation, or environmental factors like low light, confined spaces, and dust. The system is dynamically reconfigurable, functioning as a multi-level scheme and each level focuses on different types of degradation. Lower levels handle basic, fast, and resource-efficient adaptation and higher levels handle complex and computationally-demanding processes. If low-level adaptation mechanisms fail, high-level mechanisms can

intervene to support recovery. This hierarchical method enhances the resilience of state estimation, allowing the robot to effectively navigate through all forms of challenges. This approach ensures that *Super Odometry* can provide continuous, high-precision ego-motion estimation, even in the most challenging and resource-constrained environments.

Thrust #2: Developing Spatial-Temporal Scene Reconstruction for Spatial Reasoning.

To achieve more adaptive and intelligent autonomy, a dynamic, spatial-temporal 4D representation is critical for autonomous robots to reason about the evolving interactions and temporal movements of objects in 3D space. Despite pioneering efforts [13] in modeling spatial dynamics, current methods fall short of meeting real-time processing requirements. To address this limitation, we focus on online 4D scene reconstruction. Building upon our advancements in real-time 3D reconstruction and object-centric semantics (see Fig. 4), we present the *Super Scene* module, designed to achieve metric-semantic 4D reconstruction from sensor observations and poses estimated by *Super Odometry*. *Super Scene* provides an open-vocabulary, graph-structured 3D representation adaptable to varying detail levels across time. This representation supports diverse applications, from dense geometric data essential for mobility and manipulation to abstract semantic and object-level affordances critical for task planning. The system leverages 2D foundation models, integrating their outputs into 3D representations via Gaussian Splatting [14] while tracking object-centric changes over time.

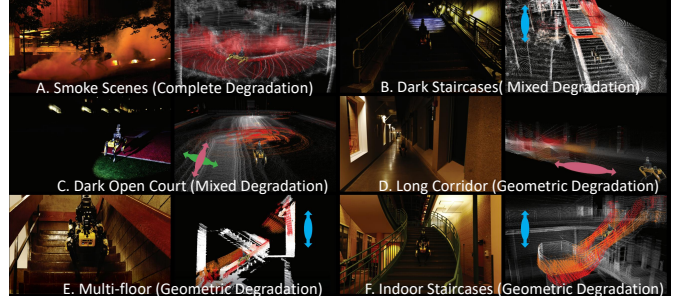


Figure 3: Our past work of robust odometry [9, 10].

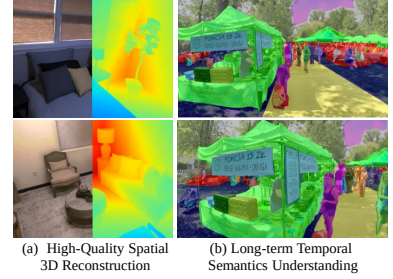


Figure 4: Our past work of spatial 3D reconstruction [11] and semantics understanding [12].

Thrust #3: Establishing Human-level Reasoning Ecosystem for Intelligent Robot Autonomy.

To achieve fully spatial autonomy, robots should reason over historical spatial-temporal observations and make informed decisions. To address this challenge, we propose *Super Reasoner*, an adaptive reasoning module that empowers Physical AI systems to make sophisticated decisions in dynamic environments efficiently. Building on our previous work in semantics reasoning under challenging conditions for temporal and spatial domains (see Fig. 5), we aim to extend these capabilities to the 4D representations generated by *Super Scene*. The core objective of *Super Reasoner* is to enable semantics-aware decision-making by integrating spatial-temporal relationships within the environment. Thus, the module performs context-sensitive reasoning by considering both the current state and the predicted evolution of the scene, with the goal of improving decision-making accuracy to achieve better autonomy.

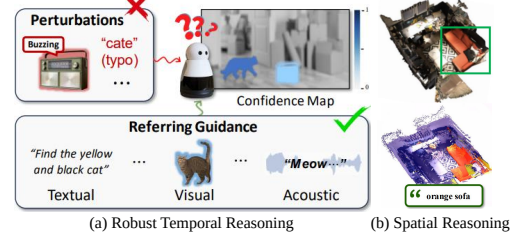


Figure 5: Our past work of robust temporal and spatial semantics reasoning [15].

Evaluation. The proposed system will undergo rigorous evaluation in both controlled and real-world scenarios. The evaluation will address three key questions: 1) *Robustness*: How well does adaptive sensor fusion perform under challenging environmental conditions such as varying light, occlusions, and dust? 2) *Accuracy*: How effectively does the 4D scene representation capture temporal and semantic data for spatial reasoning? 3) *Efficiency*: How well does the integrated system balance computational demands while achieving real-time performance on edge devices? Key performance metrics include trajectory estimation accuracy, scene reconstruction quality, computational efficiency, and semantic object retrieval precision.

One-year schedule. The project is divided into four phases: Super Odometry (Months 1–6, Zhao), Super Scene (Months 1–6, Xu), Super Reasoner (Months 7–8, Zhao&Xu), and integration (Months 9–12, Zhao&Xu).

Strength of the team. United by a shared vision of achieving spatial intelligence, Shibo’s expertise in odometry estimation and 3D reconstruction [1–3, 9, 10, 16–23] combines seamlessly with Xiaohao’s innovations in semantics understanding and reasoning [12, 15, 24–29], forming a proven partnership in spatial autonomy.

References

- [1] S. Zhao, Z. Fang, H. Li, and S. Scherer, “A robust laser-inertial odometry and mapping method for large-scale highway environments,” in *2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2019, pp. 1285–1292.
- [2] K. Ebadi, L. Bernreiter, H. Biggie, G. Catt, Y. Chang, A. Chatterjee, C. E. Denniston, S.-P. Deschênes, K. Harlow, S. Khattak *et al.*, “Present and future of slam in extreme environments: The darpa sub challenge,” *IEEE Transactions on Robotics*, 2023.
- [3] S. Zhao, Y. Gao, T. Wu, D. Singh, R. Jiang, H. Sun, M. Sarawata, Y. Qiu, W. Whittaker, I. Higgins *et al.*, “Subt-mrs dataset: Pushing slam towards all-weather environments,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 22 647–22 657.
- [4] A. Werby, C. Huang, M. Büchner, A. Valada, and W. Burgard, “Hierarchical open-vocabulary 3d scene graphs for language-grounded robot navigation,” in *First Workshop on Vision-Language Models for Navigation and Manipulation at ICRA 2024*, 2024.
- [5] J. Yang, J. Huang, Y. Chen, Y. Wang, B. Li, Y. You, A. Sharma, M. Igl, P. Karkus, D. Xu *et al.*, “Storm: Spatio-temporal reconstruction model for large-scale outdoor scenes,” *arXiv preprint arXiv:2501.00602*, 2024.
- [6] B. Chen, Z. Xu, S. Kirmani, B. Ichter, D. Sadigh, L. Guibas, and F. Xia, “Spatialvlm: Endowing vision-language models with spatial reasoning capabilities,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 14 455–14 465.
- [7] W. Yuan, J. Duan, V. Blukis, W. Pumacay, R. Krishna, A. Murali, A. Mousavian, and D. Fox, “Robopoint: A vision-language model for spatial affordance prediction for robotics,” *arXiv preprint arXiv:2406.10721*, 2024.
- [8] I. Armeni, Z.-Y. He, J. Gwak, A. R. Zamir, M. Fischer, J. Malik, and S. Savarese, “3d scene graph: A structure for unified semantics, 3d space, and camera,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 5664–5673.
- [9] S. Zhao, H. Zhang, P. Wang, L. Nogueira, and S. Scherer, “Super odometry: Imu-centric lidar-visual-inertial estimator for challenging environments,” in *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2021, pp. 8729–8736.
- [10] S. Zhao, P. Wang, H. Zhang, Z. Fang, and S. Scherer, “Tp-tio: A robust thermal-inertial odometry with deep thermalpoint,” in *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2020, pp. 4505–4512.
- [11] X. Xu, F. Xue, S. Zhao, Y. Pan, S. Scherer, and X. Huang, “Mac-ego3d: Multi-agent gaussian consensus for real-time collaborative ego-motion and photorealistic 3d reconstruction,” *arXiv preprint arXiv:2412.09723*, 2024.
- [12] X. Xu, J. Wang, X. Ming, and Y. Lu, “Towards robust video object segmentation with adaptive object calibration,” in *ACM International Conference on Multimedia (ACM MM)*, 2022.
- [13] G. Wu, T. Yi, J. Fang, L. Xie, X. Zhang, W. Wei, W. Liu, Q. Tian, and X. Wang, “4d gaussian splatting for real-time dynamic scene rendering,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 20 310–20 320.
- [14] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, “3d gaussian splatting for real-time radiance field rendering,” *ACM Trans. Graph.*, vol. 42, no. 4, pp. 139–1, 2023.
- [15] X. Li, K. Qiu, J. Wang, X. Xu, R. Singh, K. Yamazaki, H. Chen, X. Huang, and B. Raj, “Benchmarking the robustness of referring perception models under perturbations,” in *European Conference on Computer Vision (ECCV)*, 2024.
- [16] P. Wang, Z. Fang, S. Zhao, Y. Chen, M. Zhou, and S. An, “Vanishing point aided lidar-visual-inertial estimator,” in *2021 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2021, pp. 13 120–13 126.
- [17] C. Wang, D. Gao, K. Xu, J. Geng, Y. Hu, Y. Qiu, B. Li, F. Yang, B. Moon, A. Pandey *et al.*, “PyPose: A library for robot learning with physics-based optimization,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 22 024–22 034.
- [18] Z. Fang, S. Zhao, S. Wen, and Y. Zhang, “A real-time 3d perception and reconstruction system based on a 2d laser scanner,” *Journal of Sensors*, vol. 2018, no. 1, p. 2937694, 2018.
- [19] S. Zhao and Z. Fang, “Direct depth slam: Sparse geometric feature enhanced direct depth slam system for low-texture environments,” *Sensors*, vol. 18, no. 10, p. 3339, 2018.
- [20] S. Zhao, Z. Fang, and S. Wen, “A real-time handheld 3d temperature field reconstruction system,” in *2017 IEEE 7th Annual International Conference on CYBER Technology in Automation, Control, and Intelligent Systems (CYBER)*. IEEE, 2017, pp. 289–294.
- [21] Z. Fang, S. Zhao, and S. Wen, “A real-time and low-cost 3d slam system based on a continuously rotating 2d laser scanner,” in *2017 IEEE 7th Annual International Conference on CYBER Technology in Automation, Control, and Intelligent Systems (CYBER)*. IEEE, 2017, pp. 454–459.
- [22] C. Cao, L. Nogueira, H. Zhu, J. Keller, G. Best, R. Garg, D. Kohanbash, J. Maier, S. Zhao, F. Yang *et al.*, “Exploring the most sectors at the darpa subterranean challenge finals,” *Field Robotics*, 2023.
- [23] S. Scherer, V. Agrawal, G. Best, C. Cao, K. Cujic, R. Darnley, R. DeBortoli, E. Dexheimer, B. Drozd, R. Garg *et al.*, “Resilient and modular subterranean exploration with a team of roving and flying robots,” *Field Robotics*, 2022.
- [24] X. Xu, J. Wang, X. Li, and Y. Lu, “Reliable propagation-correction modulation for video object segmentation,” in *AAAI Conference on Artificial Intelligence (AAAI)*, 2022.
- [25] S. Yan, X. Xu, R. Zhang, L. Hong, W. Chen, W. Zhang, and W. Zhang, “Panovos: Bridging non-panoramic and panoramic views with transformer for video segmentation,” in *European Conference on Computer Vision (ECCV)*, 2024.
- [26] X. Li, J. Wang, X. Xu, X. Peng, R. Singh, Y. Lu, and B. Raj, “Qdformer: Towards robust audiovisual segmentation in complex environments with quantization-based semantic decomposition,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [27] W. Li, X. Xu, Y. Gu, B. Zheng, S. Gao, and Y. Wu, “Towards scalable 3d anomaly detection and localization: A benchmark via 3d anomaly synthesis and a self-supervised learning network,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [28] X. Li, J. Wang, X. Xu, X. Li, B. Raj, and Y. Lu, “Robust referring video object segmentation with cyclic structural consensus,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- [29] J. Dang, H. Zheng, X. Xu, L. Wang, and Y. Guo, “Beyond appearance: Multi-frame spatio-temporal context memory networks for efficient and robust video object segmentation,” *IEEE Transactions on Image Processing (T-IP)*, 2024.